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COURSE TITLE: OPERATING SYSTEM.

ASSIGNMENT: RESOURCE MANAGEMENT IN AN OPERATING SYSTEM.

Resource management is an essential and integral part of modern operating systems. Resource management in an Operating system can be defined as the process of managing and allocating resources such as CPU, memory, disk space, network bandwidth and Peripherals (I/O devices) to various processes and applications running on a computer system. Resource management is the dynamic allocation and de-allocation by an operating system of processor cores, memory pages, and various types of bandwidth to computations that compete for those resources. The objective is to allocate resources so as to optimize responsiveness subject to the finite resources available.

Resource management is a critical thing because resources of a computer are limited and multiple processes or users may require access to the same resources like CPU, memory, Peripherals, File storage, etc. at the same time. The operating system has to manage and ensure that all processes get the resources they need to execute, without any problems like deadlocks. Management tasks include scheduling resource use to avoid conflicts and interference between programs. Unlike most programs, which complete a task and terminate, an operating system runs indefinitely and terminates only when the computer is turned off.

There are terminologies related to the resource management in Operating systems that would help us have a grip on the concept of resource management in operating systems, they are as follows:

1. **Resource:** Resource can be anything that can be assigned dynamically or statically in the operating system. Example include CPU time, memory, disk space, peripherals and network bandwidth.
2. **Resource Allocation:** This terms defines the process of assigning the available resources to processes in the operating system. This can be done dynamically or statically.
3. **Resource Management:** It refers to how to manage resources efficiently between different processes.
4. **Process:** Process refers to any program or application that is being executed in the operating system and has its own memory space, execution state, and set of system resources.
5. **Scheduling:** It is the process of determining from multiple number of processes which process should be allocated a particular resource at a given time.
6. **Deadlock:** When two or more processes are waiting for some resource but resources are busy somewhere else and resources are also waiting for some process to complete their execution. In such condition neither resources will be freed nor process would get it and this situation is called deadlock.
7. **Semaphore:** It is the method or tool which is used to prevent race condition. Semaphore is an integer variable which is used in mutual exclusive manner by various concurrent cooperative process in order to achieve synchronization.
8. **Mutual Exclusion:** It is the technique to prevent multiple number of process to access the same resources simultaneously.
9. **Memory Management:** Memory management is a method used in the operating systems to manage operations between main memory and disk during process execution.

How an Operating System handles Resource Management

Resource management is dire because modern computer systems run multiple processes and applications simultaneously. If these processes and applications do not have sufficient resources, they can slow down, crash, or even cause the entire system to malfunction. It is extremely important for operating systems to manage resources effectively to ensure system stability and optimal performance.

Resource management includes multiplexing or sharing resources in two different ways, namely, in time and in space.

- In time multiplexed, different programs take a chance of using CPU. First one tries to use the resource, then the next one that is ready in the queue and so on. For example, sharing the printer one after another.
- In space multiplexing, Instead of the customers taking a chance, each one gets part of the resource. For example, Main memory is divided into several running programs, so each one can be resident at the same time.

Resources managed by the OS are as follows:

CENTRAL PROCESING UNIT.

One of the key resources that an operating system manages is the CPU. The CPU is responsible for executing instructions, and it is a finite resource that can only execute a limited number of instructions per second. Operating systems use various scheduling algorithms to ensure that each process gets its fair share of CPU time. Some well - known scheduling algorithms include Round-Robin, Priority-Based, and Shortest Job First.

Round Robin CPU Scheduling:

Round Robin CPU Scheduling is the most important CPU Scheduling Algorithm which is ever used in the history of CPU Scheduling Algorithms. Round Robin CPU Scheduling uses Time Quantum (TQ). The Time Quantum is something which is removed from the Burst Time and lets the chunk of process to be completed. Time Sharing is the main emphasis of the algorithm. Each step of this algorithm is carried out cyclically. The system defines a specific time slice, known as a time quantum.

First, the processes which are eligible to enter the ready queue enter the ready queue. After entering the first process in Ready Queue is executed for a Time Quantum chunk of time. After execution is complete, the process is removed from the ready queue. Even now the process requires some time to complete its execution, then the process is added to Ready Queue.

The Ready Queue does not hold processes which already present in the Ready Queue. The Ready Queue is designed in such a manner that it does not hold non unique processes. By holding same processes Redundancy of the processes increases.

After, the process execution is complete, the Ready Queue does not take the completed process for holding.

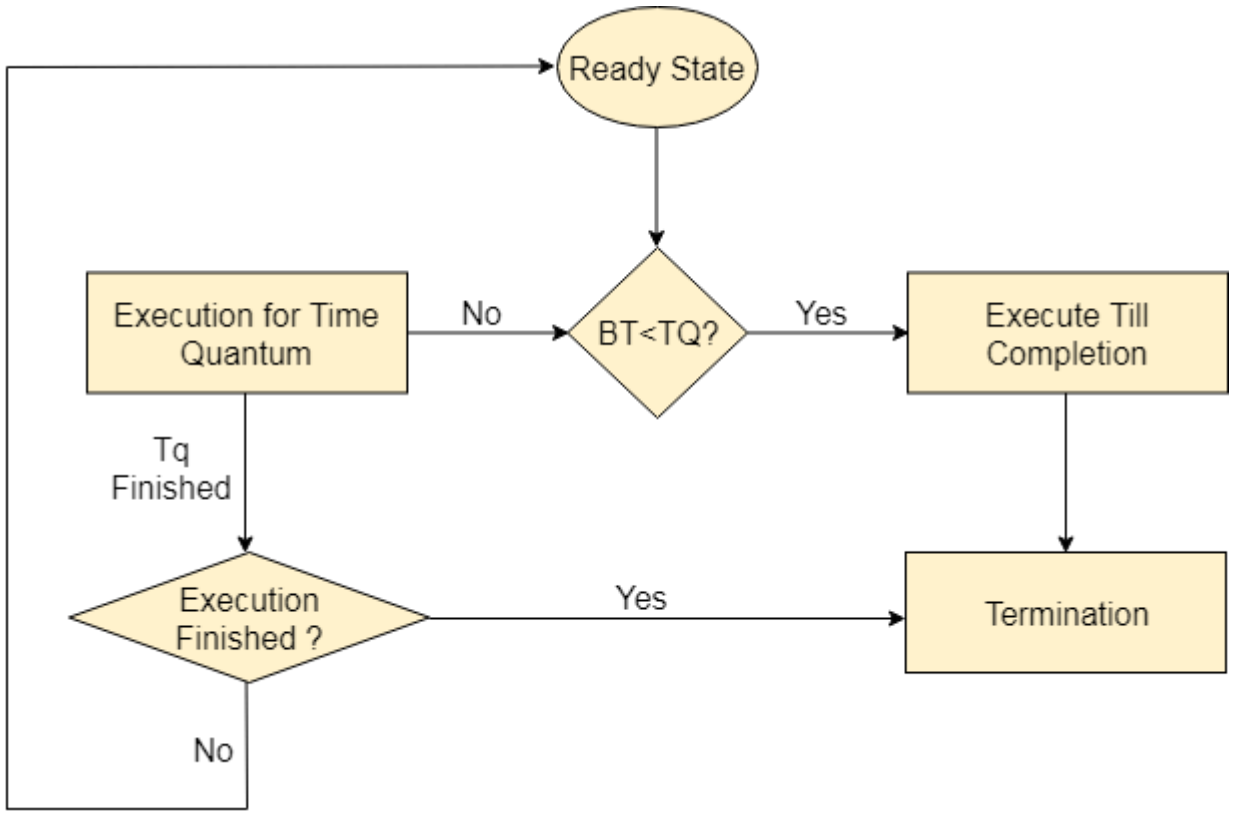

Fig. 1 Round Robin Scheduling

The Advantages of Round Robin CPU Scheduling are:

- A fair amount of CPU is allocated to each job.
- It deals with all process without any priority
- Because it doesn't depend on the burst time, it can truly be implemented in the system.
- This scheduling method does not depend upon burst time. That’s why it is easily implementable on the system.
- Once a process is executed for a specific set of the period, the process is preempted, and another process executes for that given time period.
- Allows OS to use the Context switching method to save states of preempted processes.
- It gives the best performance in terms of average response time.

Disadvantages of Round Robin CPU Scheduling:

- Low Operating System slicing times will result in decreased CPU output.
- Round Robin CPU Scheduling approach takes longer to swap contexts.
- Its performance heavily depends on time quantum.
- Priorities cannot be set for the processes.
- Round-robin scheduling doesn’t give special priority to more important tasks.
- Decreases comprehension
- Lower time quantum results in higher the context switching overhead in the system.
- Finding a correct time quantum is a quite difficult task in this system.

Priority-Based Scheduling:

Priority scheduling is a method of scheduling processes based on priority. In this method, the scheduler chooses the tasks to work as per the priority, which is different from other types of scheduling, for example, a simple round robin. Priority scheduling involves priority assignment to every process, and processes with higher priorities are carried out first, whereas tasks with equal priorities are carried out on a first-come-first-served (FCFS) or round robin basis. An example of a general-priority scheduling algorithm is the shortest-job-first (SJF) algorithm.

Priorities can be either dynamic or static. Static priorities are allocated during creation, whereas dynamic priorities are assigned depending on the behavior of the processes while in the system. To illustrate, the scheduler could favor input/output (I/O) intensive tasks, which lets expensive requests be issued as soon as possible.

Priorities may be defined internally or externally. Internally defined priorities make use of some measurable quantity to calculate the priority of a given process. In contrast, external priorities are defined using criteria beyond the operating system (OS), which can include the significance of the process, the type as well as the sum of resources being utilized for computer use, user preference, commerce, and other factors like politics, etc. Priority scheduling can be either of the following:

1. Preemptive: This type of scheduling may preempt the central processing unit (CPU) in the case the priority of the freshly arrived process is greater than those of the existing processes.
2. Non-preemptive: This type of scheduling algorithm simply places the new process at the top of the ready queue.

Indefinite blocking, otherwise called starvation, is one of the major issues concerning priority scheduling algorithms. It is a state where a process is ready to be executed but faces a long wait in getting assigned to the CPU.

It's often possible that a priority scheduling algorithm can make a low-priority process wait indefinitely. For example, in an intensely loaded system, if there are a number of higher-priority processes, the low-priority processes may never get the CPU for execution.

A remedy to starvation is aging, which is a technique used to gradually increase the priority of those processes that wait for long periods in the system.

Shortest Job First Scheduling:

Shortest Job First (SJF) scheduling algorithm schedules the processes according to their burst time.

In SJF scheduling, the process with the lowest burst time, among the list of available processes in the ready queue, is going to be scheduled next.

However, it is very difficult to predict the burst time needed for a process hence this algorithm is very difficult to implement in the system.

Advantages of SJF

1. Maximum throughput
2. Minimum average waiting and turnaround time

Disadvantages of SJF

1. May suffer with the problem of starvation
2. It is not implementable because the exact Burst time for a process can't be known in advance.

MEMORY

Another critical resource that an operating system manages is memory. Memory is the temporary storage location that holds data and instructions that a program needs to execute.

Operating systems use virtual memory techniques to manage memory efficiently. Virtual memory allows the operating system to use the hard disk as an extension of physical memory, allowing programs to use more memory than what is physically available. Virtual memory also allows the operating system to protect memory from unauthorized access and ensure that each process has its own memory space.

Every program or application is a process. The computer is always juggling hundreds of processes. When the computer is in use, you might only see just a few programs running but in reality the Operating system is handling so many processes at the same time. The OS hides the complexity so that the computer is easy to use. For example, your computer can be seen to be running few applications or programs such MS-WORD, EXCEL, COREL-DRAW, opening some documents online and watching videos. If you get to the Task Manager, under Processes Tab, you will find out that hundreds of processes are running at the same time on the system.

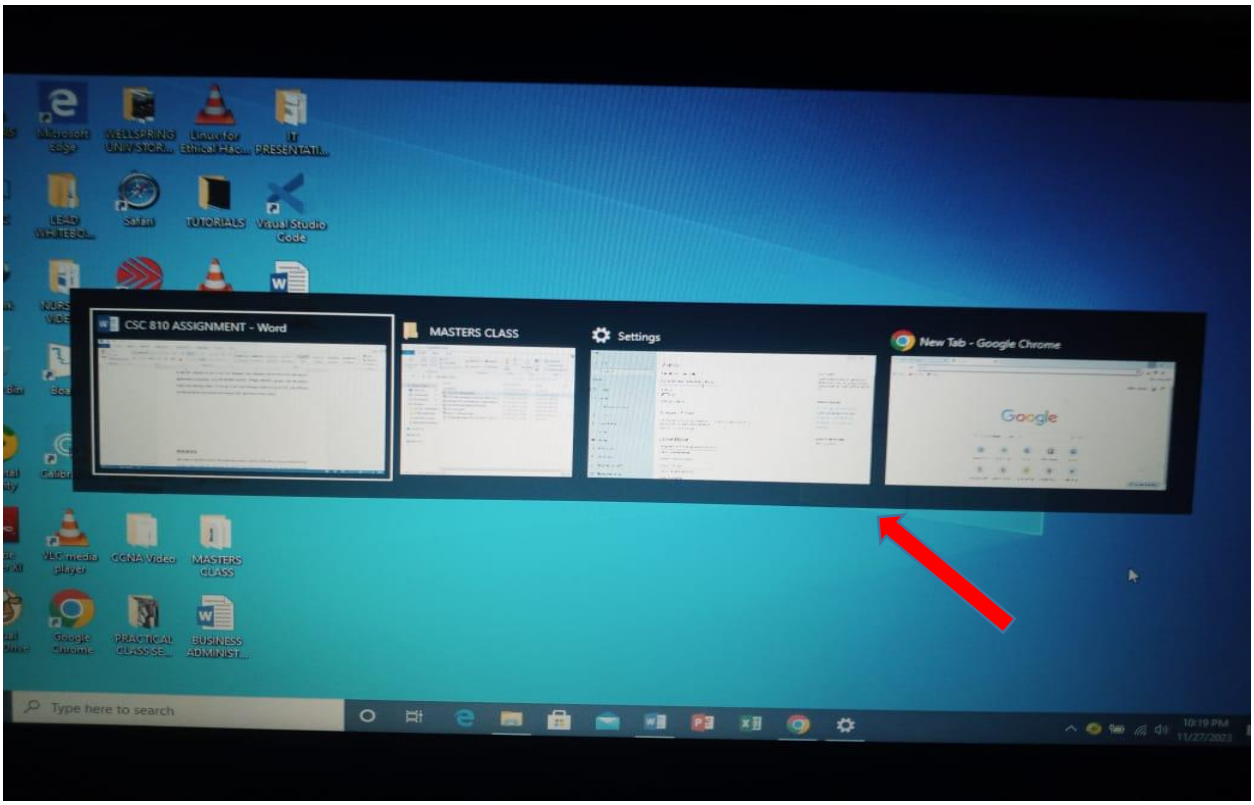

Fig.2. An image of the applications running or opened on the system

Task Manager

File Options View

Processes | Performance | App history | Startup | Users | Details | Services

Name	Status	4% CPU	35% Memory	3% Disk	0% Network	1% GPU	GPU engine	Power usage	Power usage t...
Google Chrome (6)		0%	159.7 MB	0 MB/s	0 Mbps	0%		Very low	
Microsoft Word		0%	29.3 MB	0 MB/s	0 Mbps	0%		Very low	
Settings		0%	0 MB	0 MB/s	0 Mbps	0%		Very low	
Snipping Tool		0.5%	3.0 MB	0 MB/s	0 Mbps	0%		Very low	
Task Manager		0.4%	19.6 MB	0 MB/s	0 Mbps	0%		Very low	
Windows Explorer		0.1%	38.2 MB	0.1 MB/s	0 Mbps	0%		Very low	
Background processes (59)									
360 Total Security (32 bit)		0.3%	17.5 MB	0 MB/s	0 Mbps	0%		Very low	
360 Total Security (32 bit)		0%	0.4 MB	0 MB/s	0 Mbps	0%		Very low	
360 Total Security (32 bit)		0%	55.6 MB	0 MB/s	0 Mbps	0%		Very low	
Adobe Acrobat Update Service (...		0%	0.7 MB	0 MB/s	0 Mbps	0%		Very low	
AlpsAlpine Pointing-device Driver		0%	1.9 MB	0 MB/s	0 Mbps	0%		Very low	
AlpsAlpine Pointing-device Driv...		0%	0.6 MB	0 MB/s	0 Mbps	0%		Very low	
ApMsgFwd		0.4%	1.1 MB	0 MB/s	0 Mbps	0%		Very low	
Application Frame Host		0%	5.3 MB	0 MB/s	0 Mbps	0%		Very low	
Bonjour Service		0%	1.0 MB	0 MB/s	0 Mbps	0%		Very low	
COM Surrogate		0%	2.4 MB	0.1 MB/s	0 Mbps	0%		Very low	
Component Package Support S...		0%	1.1 MB	0 MB/s	0 Mbps	0%		Very low	
Cortana (2)		0.2%	100.8 MB	0.1 MB/s	0 Mbps	0%	GPU 0 - 3D	Very low	

Fig.3. An image of the processes running on the system

WMI Provider Host (32 bit)	0%	1.6 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Windows processes (87)								
Client Server Runtime Process	0.4%	1.2 MB	0 MB/s	0 Mbps	0.4%	GPU 0 - 3D	Very low	Very low
Client Server Runtime Process	0%	0.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Console Window Host	0%	5.6 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Desktop Window Manager	1.7%	63.0 MB	0 MB/s	0 Mbps	1.5%	GPU 0 - 3D	Very low	Very low
Local Security Authority Process...	0.4%	6.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
LocalServiceNoNetworkFirewall ...	0%	8.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Registry	0%	4.6 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Application Infor...	0%	0.6 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: AVCTP service	0%	1.3 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Bluetooth Audio G...	0%	1.3 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Bluetooth Support...	0%	1.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Capability Access ...	0.1%	1.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Clipboard User Ser...	0%	4.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: COM+ Event Syste...	0%	1.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Connected Device...	0%	3.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Connected Device...	0.2%	5.8 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Service Host: Cryptographic Ser...	0%	2.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low

Fig.4. An image of the Processes running on the system

DISK SPACE

Disk space is another resource that operating systems manage. Disk space is the permanent storage location where data and programs are stored. Operating systems use file system management techniques to manage disk space efficiently. File system management allows the operating system to create, delete, and modify files and directories. File system management also ensures that data is stored and retrieved efficiently, and that disk space is allocated to each file appropriately.

Additionally, the operating system manages file systems, providing a hierarchical structure to organize and store data on storage devices. It handles file operations like creation, deletion, and modification, ensuring the integrity and security of the data.

NETWORK BANDWIDTH

Network bandwidth is another critical resource that operating systems manage. Network bandwidth is the amount of data that can be transferred over a network connection. Operating systems use various network management techniques to manage network bandwidth efficiently. Network management includes techniques such as traffic shaping, Quality of Service (QoS), and bandwidth throttling.